One-Group Delayed-Neutron Kinetics

NE 630 | Nuclear Reactor Theory

27 LESSON

Read: FNRP 5.4; one-group results from 5.5.

Relate a physical precursor concentration to (n, C), find a critical equilibrium, and infer reactivity from neutron-density traces.

1 One physical precursor

Use **bromine-87** as the only delayed-neutron precursor. Treat all remaining fission neutrons as prompt; neglect precursor capture and parent ingrowth. Here, *one group* means one precursor group.

$$^{87}\mathrm{Br} \xrightarrow{\beta^-} {}^{87}\mathrm{Kr}^* \xrightarrow{n} {}^{86}\mathrm{Kr}.$$

Use the following data:

$$\gamma = 0.01274 \text{ nuclei/fission}, \quad f_n = 0.0260,$$
$$t_{1/2} = 55.65 \text{ s}, \quad \bar{\nu} = 2.43.$$

Here f_n is the probability that a Br-87 decay produces a neutron. For the **actual** Br-87 number density $N(t)$,

$$\frac{dN}{dt} = \gamma \Sigma_f \phi - \lambda N, \qquad \lambda = \frac{\ln 2}{t_{1/2}},$$

and the delayed-neutron production rate is $f_n \lambda N$.

Br-87 decay reference: P. Dimitriou et al., *Nuclear Data Sheets* 173, 144–238 (2021), Table XXIII, ENDF/B-VIII.0 column ($t_{1/2} = 55.65$ s, $f_n = 0.026$).

2 The classic (n, C) equations

Define the delayed-neutron-equivalent concentration and delayed fraction:

$$\boxed{C = f_n N, \qquad N = \frac{C}{f_n}}, \qquad \beta = \frac{f_n \gamma}{\bar{\nu}} = 1.3631 \times 10^{-4}.$$

C counts the *expected* future delayed neutrons per unit volume; it is not the total number density of Br-87. With $\phi = \bar{\nu} n$ and $\Lambda = (\bar{\nu} \Sigma_f \bar{\nu})^{-1}$,

$$\boxed{\begin{aligned} \frac{dn}{dt} &= \frac{\rho(t) - \beta}{\Lambda} n + \lambda C + S(t), \\ \frac{dC}{dt} &= \frac{\beta}{\Lambda} n - \lambda C. \end{aligned}}$$

n, C, N have units cm^{-3}; S has units cm^{-3}s^{-1}. The reactivity is $\rho = (k-1)/k$. Identify prompt production minus loss, delayed production, and precursor loss.

3 Critical steady state, with $S = 0$

Set both derivatives to zero:

$$C_0 = \frac{\beta n_0}{\lambda \Lambda}, \qquad 0 = \frac{\rho - \beta}{\Lambda} n_0 + \lambda C_0 = \frac{\rho}{\Lambda} n_0.$$

Thus a nonzero source-free equilibrium requires $\rho = 0$. Its amplitude n_0 must be specified by the operating density or power. The corresponding physical precursor concentration is

$$\boxed{N_0 = \frac{C_0}{f_n} = \frac{\gamma \Sigma_f \phi_0}{\lambda}}.$$

Check. For $n_0 = 10^8$ cm^{-3} and $\Lambda = 5 \times 10^{-5}$ s, calculate

$C_0 =$ _____ cm^{-3}, $N_0 =$ _____ cm^{-3}.

Why can the precursor inventory greatly exceed the neutron inventory?

4 A step produces two exponential modes

Start at the critical equilibrium above. At $t = 0$, impose a constant ρ while keeping $S = 0$ and the material parameters fixed:

$$\frac{d}{dt} \begin{bmatrix} n \\ C \end{bmatrix} = \begin{bmatrix} (\rho - \beta)/\Lambda & \lambda \\ \beta/\Lambda & -\lambda \end{bmatrix} \begin{bmatrix} n \\ C \end{bmatrix}.$$

For a mode proportional to $e^{\omega t}$, the eigenvalues satisfy

$$\boxed{\Lambda \omega^2 + (\beta - \rho + \lambda \Lambda)\omega - \lambda \rho = 0.}$$

For the sub-prompt-critical insertions used below, call the larger root ω_s and the smaller root ω_f. Then

$$n(t) = A_s e^{\omega_s t} + A_f e^{\omega_f t}.$$

The initial conditions give

$$A_s + A_f = n_0, \qquad \omega_s A_s + \omega_f A_f = \frac{\rho}{\Lambda} n_0.$$

The fast mode dies away first. Both n and C remain **continuous** at $t = 0$; their derivatives can change immediately.

5 Measure the late-time period

After the fast mode has become negligible,

$$n(t) \simeq A_s e^{t/T}, \qquad \boxed{T = \frac{1}{\omega_s}}.$$

Choose two points on the straight, late part of a semilog trace:

$$\boxed{T = \frac{t_b - t_a}{\ln[n(t_b)/n(t_a)]}}.$$

$T > 0$ means growth; $T < 0$ means decay. A population doubles in $T \ln 2$ for $T > 0$, or halves in $|T| \ln 2$ for $T < 0$. Substituting $\omega = 1/T$ into the quadratic gives a direct **one-group period–reactivity relation**:

$$\boxed{\rho = \frac{\Lambda}{T} + \frac{\beta}{1 + \lambda T}}.$$

This is the one-group version of the *inhour* equation.

6 Estimate the prompt jump or drop

For a step sufficiently below prompt criticality, C changes little during the fast adjustment. Set $dn/dt \simeq 0$ while retaining $C \simeq C_0$:

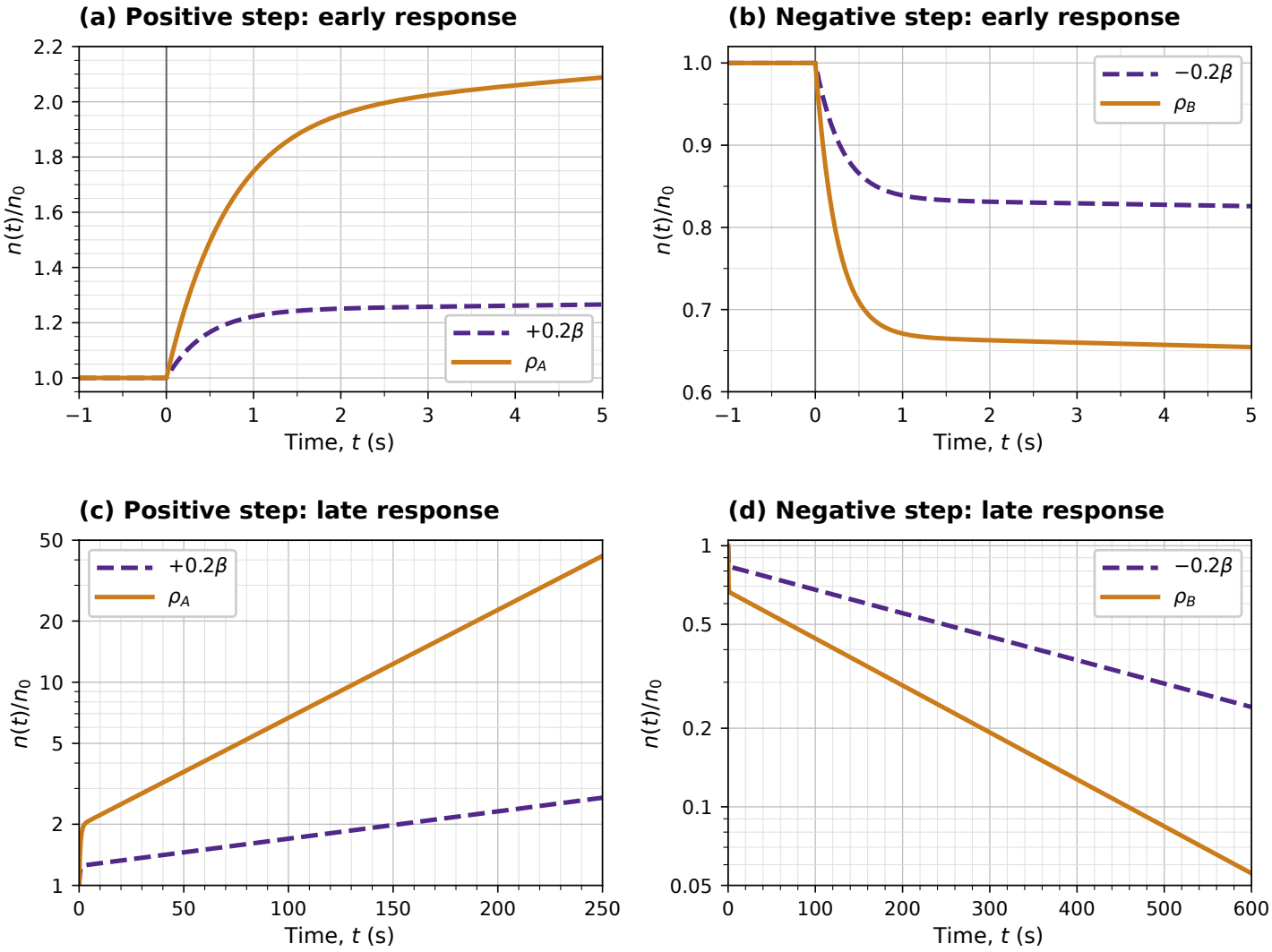
$$0 \simeq \frac{\rho - \beta}{\Lambda} n_1 + \lambda C_0 \quad \Longrightarrow \quad \boxed{J \equiv \frac{n_1}{n_0} \simeq \frac{\beta}{\beta - \rho}}.$$

Consequently, $\rho/\beta \simeq 1 - 1/J$. The apparent jump describes a short, continuous adjustment; it is not $n(0^+)/n(0^-)$. The approximation fails near $\rho = \beta$.

Units. Reactivity in dollars is ρ/β; in pcm it is $10^5 \rho$.

Graphical exercise: use a ruler and judgment

All four panels use the **same Br-87-only model**: $\beta = 1.3631 \times 10^{-4}$, $\lambda = 0.012455\ \text{s}^{-1}$, $\Lambda = 5 \times 10^{-5}$ s, $S = 0$, and initial critical equilibrium. Each reactivity step occurs at $t = 0$. The same unknown ρ_A appears in (a,c), and the same unknown ρ_B in (b,d).



1. From (a,b), estimate the apparent jump/drop factor J for each curve. Use the settled trend after the fast adjustment; extrapolate gently toward $t = 0$ if needed.
2. From the straight portions of (c,d), mark two widely separated points, estimate the *signed* period T, and show your ruler construction.
3. Estimate ρ_A/β and ρ_B/β twice: from J and from T. Check the known $\pm 0.2\beta$ curves. Explain small disagreements.

Trace	J **from plot**	T **(s) from plot**	ρ/β **from** J	ρ/β **from** T
$+0.2\beta$				
ρ_A				
-0.2β				
ρ_B				

Discuss. Why does the negative insertion not remove all neutrons immediately? Which part of each response is governed by precursor decay? Use ρ_A and ρ_B to practice conversion to dollars and pcm.